Biomedical Signal Processing and Control 79 (2023) 104186

Contents lists available at ScienceDirect

Biomedical Signal Processing and Control

journal homepage: www.elsevier.com/locate/bspc

A comprehensive analysis of dermoscopy images for melanoma detection via deep CNN features

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ARTICLE INFO

Keywords:
Skin cancer
Melanoma
Deep features
Boundary localization
CNN
Dermoscopy image

ABSTRACT

Melanoma is the fastest growing and most lethal cancer among all forms of skin cancer. Deep learning methods, mainly convolutional neural networks (CNNs) have recently brought considerable attention in detecting skin cancers from dermoscopy images. However, learning valuable features by these methods has been challenging due to the inadequate training data, inter-class similarity, and intra-class variation in the skin lesions. In addition, most of these methods demand a considerable amount of parameters to tune. To address these issues, we present an automated framework that extracts visual features from dermoscopy images using a pre-trained deep CNN model and then employs a set of classifiers to detect melanoma. Recently, few pre-trained CNN architectures have been employed to extract deep features from skin lesions. However, a comprehensive analysis of such features derived from a variety of CNN architectures has not yet been performed for melanoma classification. Therefore, in this paper, we investigate the effectiveness of deep features extracted from eight contemporary CNN models. Also, we explore the impact of boundary localization and normalization techniques on melanoma detection. The suggested approach is evaluated using four benchmark datasets: PH^2, ISIC 2016, ISIC 2017 and HAM10000. Experimental outcomes indicate that the DenseNet-121 with multi-layer perceptron (MLP) achieves a higher performance in terms of accuracy of 98.33%, 80.47%, 81.16% and 81% on PH^2, ISIC 2016, ISIC 2017 and HAM10000 datasets compared to other CNN models and state-of-the-art methods.

1. Introduction

Melanoma is the deadliest type of skin cancer and has grown exponentially across the globe. Hence, early and timely diagnosis is utmost important for preventing the severity of the disease [1]. This disease spontaneously causes black moles with an unusual border on normal skin. The mole with color variations, unstructured development of the lesion and enlargement of the lesion region also leads to melanoma [2].

In order to accurately detect melanoma, a non-invasive imaging technique, dermoscopy has been developed to aid the dermatologists in their clinical evaluation [3]. Melanoma is more lethal than other skin cancer types. Due to the advantages like good visual perception and low scoring errors, dermoscopy mages have drawn significant interest for discrimination between melanoma and benign skin lesions [4,5] compared to clinical images. Fig. 1 shows an example of melanoma and non-melanoma skin cancer images. Several classical approaches such as ABCD rule [6], 7-point checklist [7], Menzies procedure [8], and CASH [9], have been evolved to improve ability of dermatologist in distinguishing melanoma from non-melanoma images. Skin lesions are complicated to diagnose correctly even by an expert dermatologist because of the presence of inter-class similarity and intra-class variation in the skin lesions. That is, two different classes have similar types of lesions, and a particular class has diversified lesions. For example, melanoma has different types of lesions in terms of varying asymmetry, color, border, diameter, and size. Moreover, there is a high similarity between melanoma and non-melanoma skin cancer types in terms of color, size, and other features. Further, melanoma diagnosis through visual inspection is tedious, time-consuming and expensive. Hence, there is a strong need to develop automated methods that enhance the melanoma diagnosis performance [10,11].

Many algorithms have been proposed for automated analysis of dermoscopic images. An overview of related work over the past decades can be found in [12,13]. The earlier systems predominantly use a traditional pipeline of the following stages [14]: (i) pre-processing, (ii) lesion segmentation, (iii) feature extraction, (iv) feature selection (optional) to eliminate irrelevant, redundant, or noisy features, and (v) classification. Feature extraction and classification are the two crucial stages and have shown the most variations. Features like color, texture, border, shape-related descriptors and classifiers such as support vector

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https://doi.org/10.1016/j.bspc.2022.104186
Received 10 September 2021; Received in revised form 11 August 2022; Accepted 4 September 2022
Available online 18 September 2022
1746-8094/

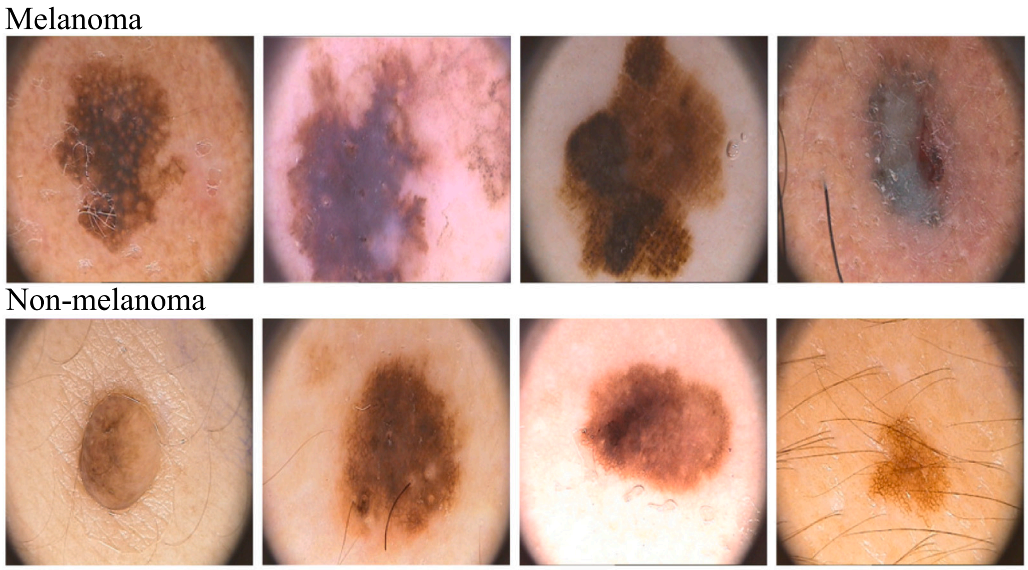

Fig. 1. Sample of original dermoscopy images of skin lesions from melanoma and non-melanoma category.

machine, feed-forward neural network, random forest, naive Bayes, etc., have been extensively employed [15,16]. However, finding an appropriate feature, feature extractor and classifier has remained a complicated task. The earlier systems used hand-crafted or manual feature extraction techniques. Extracting the set of features from many images using a manual process is time-consuming and the feature set is also minimal. Moreover, the performance achieved by these methods is still not satisfactory because of the inter-class uniformity and intra-class diversity among the skin lesion classes, which may not be suitable for clinical practice.

On the other hand, deep learning methods, especially convolutional neural networks (CNNs) have achieved astounding success in many image recognition tasks in the past few years [17–20]. The major characteristic of CNN is that it facilitates the extraction of detailed visual representation from given training data [21]. The recent literature reveals that the CNN models pre-trained on a large-scale ImageNet dataset [22] also produced an encouraging performance for other image recognition tasks and do not require retraining. Hence, few investigations have been performed for dermoscopy image classification using transferred CNN features [23–25]. Further, for skin images with inter-class uniformity and intra-class diversity, the classification of melanoma using deep features derived directly from original images may lead to a poor performance [26–28]. Hence, a proper ROI extraction method could be harnessed which can only capture the features from the essential part of the skin lesion.

In general, deep learning techniques learn relevant features when trained with a huge amount of data. However, the size of the publicly available skin cancer dataset is insufficient to train such deep learning models. In addition, these models require enormous parameters, which may lead to overfitting while working with limited data. Further, many earlier CNN models extract features directly from original images that may ignore the important visual features of the lesion region. To handle these issues, we extract the region of interest (ROIs) by introducing a boundary localization technique and present a framework for automatic and accurate melanoma recognition in dermoscopy images, where we use a deep CNN model pre-trained with a large-scale dataset to extract prominent features followed by a set of classifiers for melanoma classification. Limited work has been reported to date using such methods. To the best of our knowledge, the efficacy of a variety of contemporary pre-trained CNN models has not yet been fully investigated for skin cancer detection.

The main contributions of this paper are as follows:

- We design an automated framework using different pre-trained deep CNN features for effective melanoma classification in dermoscopy images.
- We investigate the effectiveness of deep visual features extracted from eight different types of pre-trained CNN architectures and compare the performance among them with a set of standard classifiers.
- We validate various possible combinations of CNN models and classifiers using four benchmark datasets such as PH^2 [29], ISIC 2016 [30], ISIC 2017 [31] and HAM10000 [32]. Also, we compare the performance of the best performing model with state-of-the-art methods.
- We perform ablation studies to demonstrate the influence of several crucial factors such as prepossessing, train-test split ratio, normalization technique, and features at different CNN layers on our suggested approach. Further, we verify the performance of the model with fused deep features concatenated from three different levels of the CNN architecture such as data-level, intermediate-level, and decision-level.

The remainder of the manuscript is structured as follows: Section 2 outlines a summary of the related work. The proposed methodology is presented in Section 3, with detailed architecture. Section 4 explained about detail of the experimental settings. The results and analysis are discussed in Section 5, with ablation studies and comparisons with existing techniques. Finally, in Section 6, we conclude our manuscript.

2. Related work

Numerous algorithms have been proposed for automated detection of melanoma using dermoscopy images over the last few years. A detailed review of the various feature extraction techniques has been presented in [14]. Recently, a study on melanoma identification and recognition has been reported in [33]. Earlier automatic methods were primarily based on manual feature engineering and classification. On the other hand, there has been a rise in applying deep learning approaches for automated skin image analysis in recent years. An overview of these methods is given below. Further, a combination of deep and hand-engineered features has been explored in [34] to improve the detection performance.

Table 1
Summary of deep features based skin cancer detection methods using dermoscopy images.

Author	Year	Method	Network	Dataset	Classes
Gessert et al. [44]	2019	CNN + Patch-Based Attention	Inception-V3	HAM10000	7
Zhang et al. [45]	2019	ARL-CNN	ResNet	ISIC 2017	2
Razzak et al. [46]	2020	Multistage Fully Connected Residual Network	FcResNet	HAM10000	7
Jayapriya et al. [47]	2020	Hybrid fully convolutional networks	VGG16, GoogLeNet	ISIC 2016, 2017	2
Wu et al. [48]	2020	ARDT-DenseNet	DenseNet-100	ISIC 2016, 2017	2
Zgha1 et al. [49]	2020	Sparse autoencoder + SVM	Autoencoder	PH2	3
Tang et al. [50]	2020	GP-CNN-DTEL	Xception	ISIC 2016 and 2017	2
Chaturvedi et al. [51]	2020	CNN Ensemble model	InceptionV3, ResNeXt101, InceptionResNetV2, Xception and NASNetLarge	HAM10000	7
Rajput et al. [52]	2021	Customized AlexNet with new activation fuction	AlexNet	HAM10000	7
Pacheco et al. [53]	2021	Attention-Based Mechanism + MetaBlock	EfficientNet-B4, DenseNet-121, MobileNet-v2, ResNet-50 and VGG-13	ISIC 2019 and PAD-UFES-20	8 and 6
Abdar et al. [54]	2021	Uncertainty quantification using Ensemble CNN models	ResNet152, DenseNet-201, InceptionResNet and MobileNet	ISIC 2019 and Kaggle skin cancer dataset	2
Gajera et al. [55]	2021	Deep CNN Features extraction	MobileNet	PH2	2
Khan et. al. [56]	2021	Deep Learning Features and Improved Moth Flame Optimization	ResNet101and DenseNet-201	HAM10000	7
Khan et. al. [57]	2021	Intelligent fusion-assisted for smart healthcare	30-layered CNN framework	HAM10000 and ISIC 2019	7 and 8
Arshad et al. [58]	2021	Computer-Aided Diagnosis System	ResNet-50, ResNet-101	HAM10000	7
Gajera et al. [59]	2022	Patch-based local deep feature extraction	VGG16, VGG19, Inception-V3, ResNet-50, DenseNet-121, MobileNet and EfficientNet-B0	ISIC 2016 and 2017	2

2.1. Methods based on hand-engineered features

The rule “ABCD” has gained more popularity in classifying skin images into benign or melanoma [6]. A method employing highlighted color and texture features was proposed in [35] to classify melanocytic lesions from non-melanocytic lesions. Ganster et al. [36] used hand-designed features like shape, border, and color descriptors along with feature optimization framework and KNN to classify melanoma from benign lesions. A method with AdaBoost classifier and 57 different hand-engineered features including border, texture, color and structure features was proposed in [37]. Barata et al. [38] designed texture and color based features for the classification of skin lesions. In [39], lesion regions were first generated by the self-generating neural network and the features like color, texture, and border were utilized for training the neural network ensemble model for melanoma classification. Shape and symmetrical feature based method has been proposed in [40] for detection of melanoma. Ozkan et al. [41] derived features such as asymmetry, color, and vascular structure from dermoscopy images and used four different classifiers for skin lesion classification. In [42], features like geometrical, asymmetry, color, and texture descriptors were derived and fed to different classifiers such as KNN, logistic regression and decision tree to classify melanocytic lesions. The use of a gray-level co-occurrence matrix (GLCM) was also provided. Bi et al. [43] extracted scale invariant feature transform (SIFT) features and employed the joint reverse classification (JRC) method for automatic melanoma detection.

2.2. Methods based on deep features

Recently, CNN has become a method of choice for skin cancer detection in dermoscopy images. To our knowledge, the first work that employed deep learning over dermoscopy images was published in 2015 [60] and thereafter, many papers have been published based on deep learning to learn effective visual representations. Demyanov et al. [61] developed a five-layer CNN architecture to distinguish two types of dermoscopic patterns. Yu et al. [62] proposed a deep residual network (DRN) to classify melanoma from skin lesion images. A single deep network based on Inception V3 architecture was designed [63] for automated classification of skin cancer. A hybrid scheme for automated melanoma recognition was proposed in [64] where features were extracted using the fine-tuned deep residual network and then aggregated using fisher vector. Finally, classification was performed using SVM. To detect melanoma, an ImageNet pre-trained CNN model (AlexNet) was used to derive high-level feature representations. Maia et al. [65] derived the features using pre-trained CNN architectures like VGG, ResNet-50 and Inception V3, and compared their performances using various classifiers. A multi-class skin disease classification method was proposed in [66] using AlexNet based deep CNN features and SVM. Recently, a fully connected multistage deep residual CNN was developed in [46] for accurate detection of melanoma. A CNN with a patch-based attention module has been proposed in [44] for skin lesion classification. Jayapriya et al. [47] suggested deep features combined with a handmade features-based melanoma detection technique. Wu et al. [48] developed a densely connected CNN with attention residual learning to classify skin lesions. In [49], a melanoma classification approach based on sparse auto-encoder and SVM has been described. Several research [45,50] built end-to-end deep CNN models for skin cancer detection and obtained substantial results. Chaturvedi et al. [51] created a CNN-based ensemble technique for multi-class skin cancer classification. In [52], a modified AlexNet architecture with a novel activation mechanism has been proposed for accurate skin cancer classification. An attention-based method with a metadata processing block has been proposed for skin cancer classification in [53]. Abdar et al. [54] suggested a methodology for skin cancer classification based on uncertainty quantification. Deep features have been extracted from dermoscopy images using MobileNet architecture in our earlier study [55], and SVM has been applied to increase the performance of melanoma identification. In [56] skin lesion segmentation and classification perform using in-depth features and enhanced moth flame optimization. An intelligent fusion-assisted smart healthcare system has been proposed in [57] for skin lesion localization and classification with different layers of CNN networks. A two-stream deep neural network information fusion architecture for multi-class skin cancer classification has been proposed in [67]. Arshad et al. [58] presented a multi-class skin lesion classification framework using ResNet-50, ResNet-101 with augmented HAM10000 dataset. Recently, Gajera et al. [59] proposed a patch-based local deep feature extraction method using different pre-trained CNN models and kernel PCA for skin cancer classification. Table 1 shows a summary of existing deep features based methods for skin cancer classification.

2.3. Cutting edges and gaps

We discussed two kinds of feature extraction methods in Sections 2.1 and 2.2. The hand-engineered feature descriptors such as shape, border, color, and texture have been extracted manually, which is time-consuming. Further, a set of machine learning techniques have been used to perform classification. However, finding an appropriate feature extractor and classifier has remained a complicated task. On the other hand, deep learning-based approaches, particularly CNNs, learn high-level features and do not require manual hand-engineered features, achieving better performance. Although these techniques always need a high amount of training data for effective feature learning, the use of pre-trained CNN architectures via transfer learning concept tends to address such issues.

Several challenges still exist in accurately classifying skin lesions: (1) How can the best features be extracted automatically from skin images to improve the classification performance? (2) Can a feature fusion approach lead to better performance? (3) Can a normalization technique be used to maintain stability in the network? (4) Which CNN model is more appropriate for feature extraction from skin lesions? (5) How to expand the training data size (use of external data or/and data augmentation techniques)? (6) How to design an end-to-end learning model with limited training data? In this work, we investigated different CNN models for automated feature extraction with image prepossessing and different train-test split ratios to find the best suitable feature extractor from skin lesions. In addition, the impact of several normalization techniques and feature fusion approaches have been investigated.

3. Proposed methodology

In this section, we present our proposed model which comprises three stages: preprocessing, deep feature extraction, and classification as shown in Fig. 2. A detailed description of each stage is given below.

3.1. Image preprocessing

(1) Boundary Localization: Due to the usage of electronic dermatoscope, dermoscopy images may influence by the noise that affects the quality of images. It contains foreign objects such as air bubbles, skin lines, and hair. The skin lesion area covers a small portion of the image and other regions carry natural skin tissues that are unimportant but may affect the classification performance.

To address this issue, we use the boundary localization technique that removes unnecessary regions and extracts essential parts (ROIs) of the lesion. Also, it helps in extracting the required image features. The entire procedure has been listed step-wise in Algorithm 1.

(2) Image Resize and Normalization: Boundary localization results in images of different dimensions and to make CNN architectures computationally feasible, the images have been resized. The resultant images are shown in Fig. 3.

Thereafter, normalization is employed over the images to limit each feature within a uniform range. In addition, it enhances computational efficiency and establishes stability in the network. The impact of four different image normalization approaches has been studied which are defined as follows:

(I) All-image-normalization (denoted as all-image-norm): Images are normalized by subtracting the mean pixel value and dividing by the standard deviation computed over the entire dataset. The procedure is mathematically stated as follows.

$$\mu_{ain} \leftarrow \frac{1}{n*m}\sum_{j=1}^{n}\sum_{i=1}^{m} I_i^j(r,c,C_h) \tag{1}$$

$$\sigma_{ain}^2 \leftarrow \frac{1}{n*m}\sum_{j=1}^{n}\sum_{i=1}^{m} [I_i^j(r,c,C_h)-\mu_{ain}]^2 \tag{2}$$

Algorithm 1: Boundary localization

Input: RGB images and binary masks
Output: Images contains skin lesion (ROIs)

for *all images* **do**
  $originalimage \leftarrow imageread()$ // Read images
  $mask \leftarrow imageread()$ // Read mask
  $gl \leftarrow grayscale(mask)$ // Convert into gray
  $gray \leftarrow gaussianblur(gl)$ // Blur image
  $thresh \leftarrow findthreshold(gray, pixelVal, maxVal, threshTech)$ // Find threshold using given pixel value
  $cnt \leftarrow findcontour(thresh, retrievalMode, apprxMethod)$ // Retrieves contours from the image
  $cntmax \leftarrow max(cnt)$ // Store max value
  for *value in cntmax* **do**
    // Apply rectangle and crop
    $<x,y,w,h> \leftarrow boundingRect(value)$ // (x,y) be the coordinate of the rectangle and (w,h) be its width and height
    $ROI \leftarrow originalimage[y:y+h, x:x+w]$ // Apply coordinate on original image
  end
  return ROI
end

$$\hat{I}_{new} \leftarrow \frac{I_{old}(r,c,C_h)-\mu_{ain}}{\sqrt{\sigma_{ain}^2}} \tag{3}$$

where, μ_{ain} represents the mean pixel value computed over the whole training samples, $I(r,c,C_h)$ represents the image with r as rows, c as columns and C_h as channels, m denotes the total number of pixels in an image, n represents the total number of training images, σ_{ain}^2 denotes the variance and $\hat{I}_{new}$ denotes the normalized image.

(II) Per-image-normalization (denoted as per-image-norm): In this approach, images are normalized by subtracting the mean pixel value and dividing by the standard deviation computed over a single image, which is inspired by Yu et al. [68]. The procedure is mathematically expressed as follows.

$$\mu_{pin} \leftarrow \frac{1}{m}\sum_{i=1}^{m} I_i(r,c,C_h) \tag{4}$$

$$\sigma_{pin}^2 \leftarrow \frac{1}{m}\sum_{i=1}^{m}[I_i(r,c,C_h)-\mu_{pin}]^2 \tag{5}$$

$$\hat{I}_{new} \leftarrow \frac{I_{old}(r,c,C_h)-\mu_{pin}}{\sqrt{\sigma_{pin}^2}} \tag{6}$$

where, μ_{pin} represents the mean pixel value computed over each single training sample, σ_{pin}^2 denotes the variance and $\hat{I}_{new}$ denotes the normalized image.

(III) Per-channel-normalization (denoted as per-channel-norm): The images are normalized by subtracting the mean intensity values and dividing by standard deviation calculated across individual channels of the entire dataset which is mathematically defined as follows.

$$\mu_r \leftarrow \frac{1}{n*r*c}\sum_{j=1}^{n}\sum_{i=1}^{r,c} I_i^j(r,c,C_r) \tag{7}$$

$$\sigma_r^2 \leftarrow \frac{1}{n*r*c}\sum_{j=1}^{n}\sum_{i=1}^{r,c}[I_i^j(r,c,C_r)-\mu_r]^2 \tag{8}$$

$$\hat{I}_{new} \leftarrow \frac{I_{old}-\mu_{r,g,b}}{\sqrt{\sigma_{r,g,b}^2}} \tag{9}$$

where, μ_r represents the mean pixel value computed over red channel intensity valued of whole training samples, r denote as rows, c

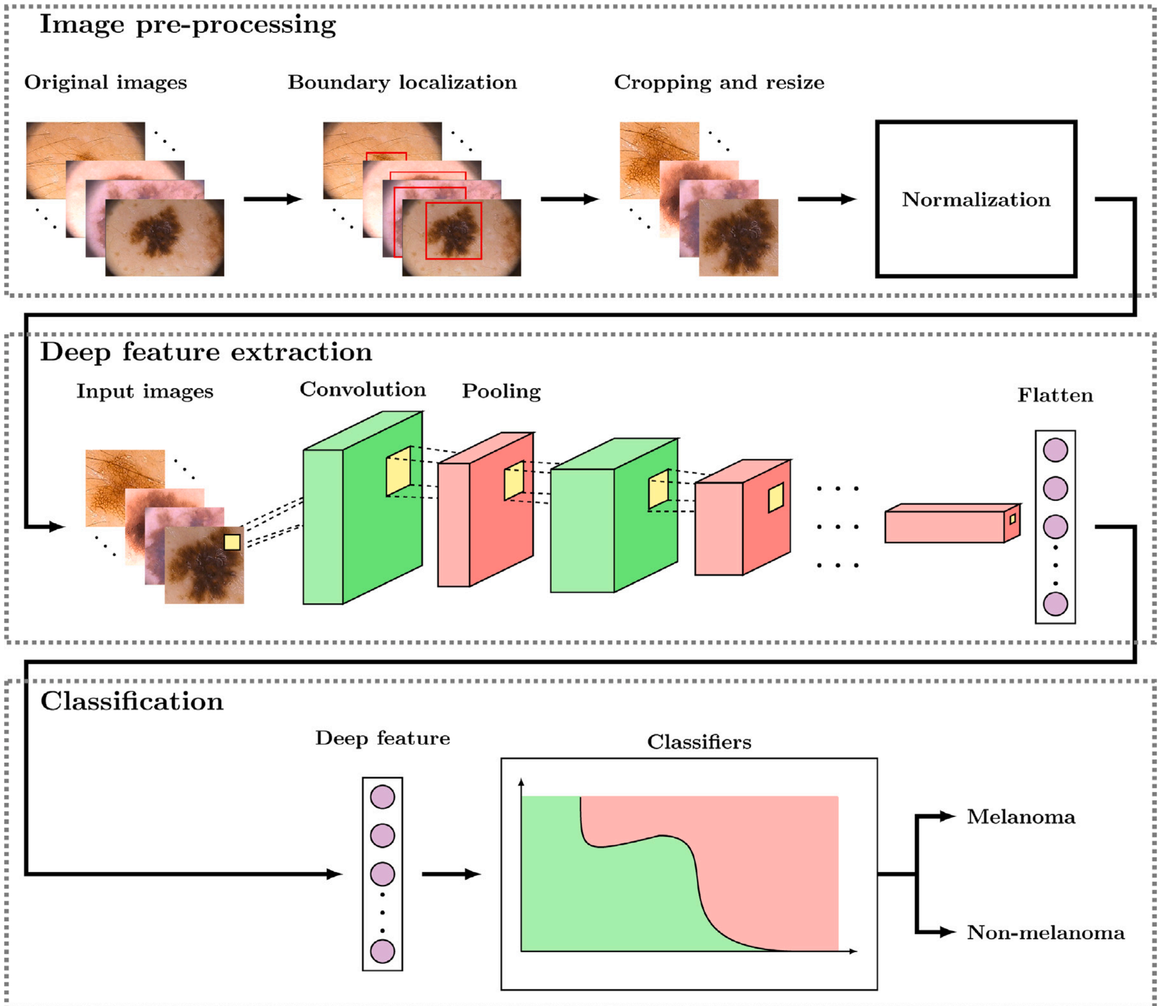

Fig. 2. Architecture of the proposed model for melanoma detection.

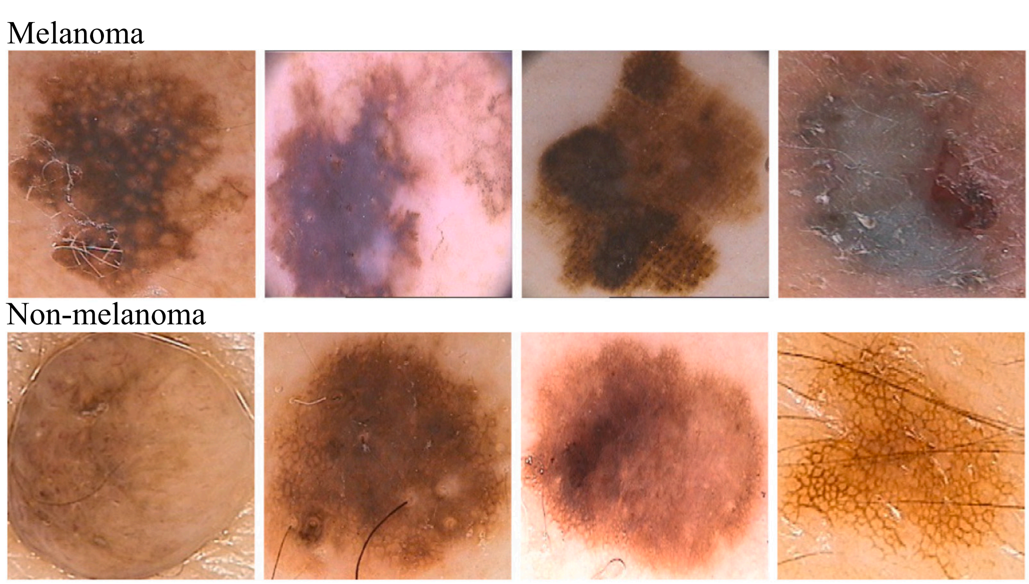

Fig. 3. Results of pre-processing (boundary localization and resize).

Table 2
Details of deep features extracted from pre-trained CNN architectures.

Architecture	Layer	Number of features
AlexNet	FC Layer 2	4096
VGG-16	FC Layer 2	4096
VGG-19	FC Layer 2	4096
Inception V3	GAP Layer	2048
ResNet-50	GAP Layer	2048
DenseNet-121	GAP Layer	1024
MobileNet	GAP Layer	1024
EfficientNet B0	GAP Layer	1280

as columns and C_r as red channel image, σ_r^2 denotes the variance computer over the red channel and $\hat{I}_{new}$ denotes normalized images.

(IV) Max-pixel-normalization (denoted as max-pixel-norm): The images are normalized by dividing each pixel value by 255 which is expressed as follows:

$$\hat{I}_{new} \leftarrow \frac{I_{old}(r, c, C_h)}{255.0} \tag{10}$$

3.2. Deep feature extraction

Deep learning algorithms, in particular, CNN have become more prevalent for a variety of computer vision tasks, including skin cancer classification. However, most of the CNN architectures not only demand a huge amount of parameters but also require large-scale datasets for effective training. Since the size of the benchmark datasets available for skin cancer is very limited, training deep CNN models with enormous parameters would be very challenging. Also, it may lead to overfitting. To alleviate these issues, transfer learning concept has been widely adopted in CNNs. Traditional CNN layers are connected in a cascade manner as stated in (11), which makes it difficult for the network to expand deeper and broader, thereby leading to a vanishing gradient problem. ResNet handles this problem by introducing shortcut connections as expressed in (12).

$$a_l = C_l(a_{l-1}) \tag{11}$$

$$a_l = C_l(a_{l-1}) + a_{l-1} \tag{12}$$

$$a_l = C_l([a_0, a_1, \dots, a_{l-1}]) \tag{13}$$

where, l denotes the layer index and C denotes the non-linear operation. The features of the lth layer is represented by a_l.

However, we adopt DenseNet-121 [19] architecture to extract meaningful visual representations from skin lesions. DenseNet mainly consists of dense blocks which utilize dense connections between the layers to concatenate feature-maps progressively from all previous layers as described in (13) and thereby, increasing the feature learning ability. DenseNet offers several benefits: it facilitates feature reuse, it strengthens feature flow, and it reduces the problem of either bursting or gradient vanishing. Further, DenseNet requires relatively fewer parameters than many popular CNN architectures such as ResNet, VGG, and AlexNet. Therefore, densely connected convolutional networks have succeeded in many computer vision applications compared to other networks. Wang et al. [69] have investigated different versions of DenseNet architecture, such as DenseNet-121, DenseNet-169, and DenseNet-201, using the transfer-learning-based approach. Recently, zhu et al. [70] proposed a DenseNet-Based SNN (DSNN) model for explainable disease classification.

Moreover, a comprehensive analysis of deep features extracted using contemporary CNN models has not yet been investigated thoroughly. Hence, we explored the effectiveness of various deep features derived using pre-trained CNN architectures such as AlexNet [71], VGG-16 [17], VGG-19 [17], Inception V3 [72], ResNet-50 [18], MobileNet [73], and EfficientNet B0 [74] for melanoma classification. These architectures are the most popular pre-trained models and are winners/runners-up of the ImageNet challenge. The recent literature has witnessed the application of few pre-trained CNN architectures for the extraction of relevant features [23]. Also, a comparative analysis among these models has been performed.

The features have been extracted from the second last layer of different architectures. In particular, fully connected (FC) layer for AlexNet and VGG and global average pooling (GAP) for remaining architectures. The details of the feature extraction layer and the number of features for considered architectures are shown in Table 2. We also investigated the effect of features derived from different layers of the CNN architectures and applied the features fusion approach to compare their performance. The results are illustrated in Section 5.

3.3. Classification

We employed an MLP [75] classifier with four hidden layers of varying sizes to perform the classification, as shown in Fig. 4. The network is configured as $I_n \times h_n^{(1)} \times h_n^{(2)} \times h_n^{(3)} \times h_n^{(4)} \times C_n^{(o)}$, where I_n denotes the input features, $h_n^{(1)}$ represents the neurons in the first hidden layer with 50 neurons, $h_n^{(2)}$ and $h_n^{(3)}$ represent the neurons in the second and third hidden layers with 200 neurons, $h_n^{(4)}$ represents the fourth hidden layer with 100 neurons. The output class prediction is represented by $C_n^{(o)}$. The Adam optimization approach has been used, which combines RMSprop, stochastic gradient descent, and momentum. The other hyperparameters of the MLP classifier were set experimentally to achieve higher performance. The advantages of employing an MLP classifier are, (1) MLP can learn and model non-linear and complex relationships. (2) MLP does not impose any restrictions on the input variables. (3) It improves the generalization ability.

The efficacy of MLP classifier is compared with other popular set of classifiers such as logistic regression (LR) [76], support vector machine (SVM) with linear and RBF kernel [77], AdaBoost (AB) [78], random forest (RF) [79], linear discriminant analysis (LDA) [80], K-nearest neighbors (KNN) [81], decision tree (DT) [82], and Naive Bayes (NB) [83]. The results are shown in Section 5.

4. Experimental setting

4.1. Dataset

The proposed method was evaluated using four benchmark datasets: PH^2 [29], ISIC 2016 [30], ISIC 2017 [31] and HAM10000 [32].

PH^2 [29] dataset has a total of 200 dermoscopic images of melanocytic lesions (80 normal nevi, 80 atypical nevi and 40 melanoma). However, it has been divided into two classes: melanoma and non-melanoma by merging two types of common and atypical nevi.

The international skin imaging collaboration (ISIC 2016) [30] dataset comprises 900 images for training and 379 images for testing. In particular, the training set contains 727 non-melanoma and 173 melanoma cases, whereas the testing set carries 304 non-melanoma and 75 melanoma cases. The size of the images varies from 556 × 679 to 2848 × 4828 pixels. The main task of this dataset is melanoma classification.

The ISIC 2017 [31] skin lesion classification dataset contains 2000 training and 600 testing images. The dataset has 374 melanoma cases, 1372 benign nevi cases, and 254 seborrheic keratosis instances. We created a non-melanoma class by combining benign nevi and seborrheic keratosis instances. The HAM10000 [32] dataset contains images of seven distinct types of skin lesions: Actinic Keratoses and Intraepithelial Carcinomae (AKIEC), Basal Cell Carcinoma (BCC), Benign Keratosis-like Lesions (BKL), Dermatofibroma (DF), Melanoma (MEL), Melanocytic Nevus (NV) and Vascular Lesion (VASC). There are 10015 images in the collection. However, the dataset is imbalanced, with few images in some classes.

We randomly split the datasets using 70:30 ratio for training and testing, respectively. The performance of the models was also investigated using different splitting ratios like 50:50, 60:40, 80:20 ratio as well as 5-fold cross-validation.

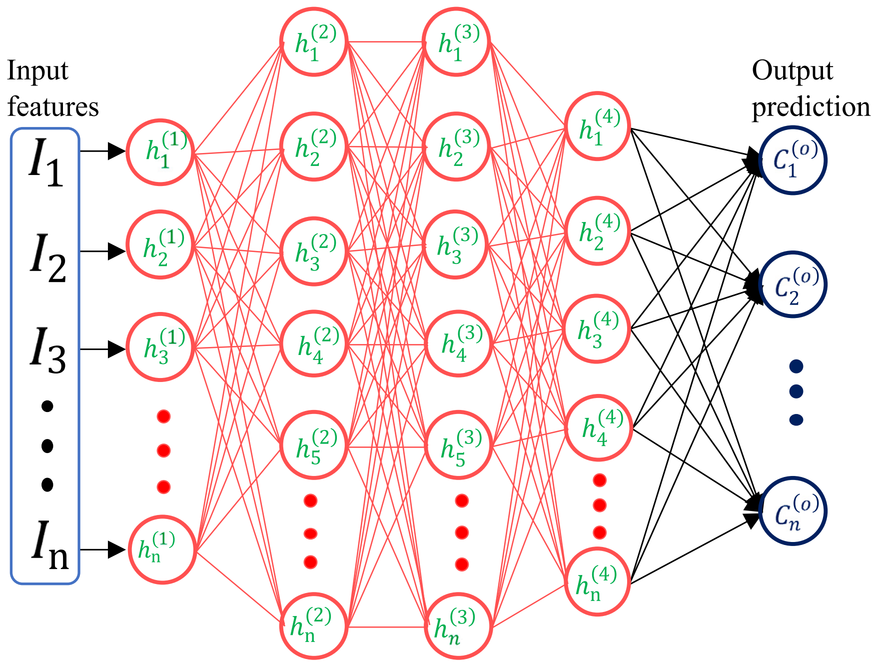

Fig. 4. The architecture of MLP classifier.

4.2. Evaluation metrics

The performance metrics used for evaluation of melanoma classification include accuracy (Acc), precision (Pre), sensitivity (Sen), specificity (Spec), and F1-score(F1). The important parameters used by these metrics are (1) True Positive (TP): when the image belongs to the melanoma class and is predicted as melanoma, (2) True Negative (TN): when image belongs to the non-melanoma class and predicted as non-melanoma, (3) False Positive (FP): when image belongs to a non-melanoma class and predicted as melanoma, and (4) False Negative (FN): when image belongs to melanoma class and predicted as non-melanoma.

4.3. Implementation detail

We implemented the proposed approach using Python based Keras [84] and Sklearn [85] library. Experiments were conducted on a platform with a configuration of Intel Core i7-7700 (8) CPU @ 4.20 GHz and 16 GB RAM with a single NVIDIA GeForce GTX 1050Ti GPU. For feature extraction, several contemporary architectures pre-trained with ImageNet were used which take an image resolution of 224 × 224 or 299 × 299 as an input. For classification, various classifiers were employed whose parameters were determined empirically.

5. Results and discussion

5.1. Evaluation of different CNN architectures with classifiers

The features extracted from the spatial exploitation based CNNs like AlexNet, VGG-16 and VGG-19, the depth based CNNs like Inception V3 and MobileNet, and the multi-path based CNNs like ResNet-50, DenseNet-121 and EfficientNet B0 are fed to a set of classifiers and the results on PH^2 dataset are shown in Table 3. The results were computed under a similar experimental setup to derive a fair comparison. It is worth mentioning that we adopted per-channel-norm normalization technique and 70:30 train-test split ratio for this experiment.

It is observed that DenseNet-121 achieved higher classification performance with 98.33% of accuracy, 100.00% of sensitivity, 97.91% of specificity and 96.00% of F1-score compared to other combinations. A model with 100% sensitivity manifests that all patients with melanoma are correctly identified. Results also show that the classifier MLP obtained better performance than other classifiers with a majority of the networks. It can also be noticed that a comparable performance was achieved with both VGG and Inception V3 architectures which obtained an accuracy of 95.00%.

5.2. Ablation studies

In this section, we performed ablation studies to show the impact of several crucial factors such as prepossessing, train-test split ratio, normalization technique, and features at different CNN layers on our suggested approach. It is worth noting that all these experiments were performed using PH^2 dataset and the factors that yield the best performance were considered for evaluation of other datasets.

5.2.1. Experiments with and without preprocessing

The objective of this experiment is to demonstrate the effectiveness of the ROI extraction procedure for correct classification of melanoma. To verify this, we assessed the efficiency of the best performing model (i.e., DenseNet-121+MLP) on the original images (without preprocessing) as well as preprocessed images of PH^2 dataset and the results are given in Table 4. We adopted the per-channel-norm normalization method and 70:30 train-test ratio for this experiment.

It is observed that the proposed approach with preprocessing yielded better melanoma classification performance than the approach without preprocessing. An accuracy of 98.33% and 95.00% was achieved with and without preprocessing, respectively. The observation also demonstrates that the accuracy, precision, sensitivity and F1-score are increased by ~3.33%, ~1.40%, ~16.67%, and ~9.05%, respectively; however, the specificity values are found similar for both the scenarios.

5.2.2. Experiments on different train and test splitting ratios

In this experiment, we test the performance of the model with different split ratios such as 80:20, 70:30, 60:40, and 50:50 as well as 5-fold cross-validation (CV). We considered three best performing CNN architectures (i.e., VGG-16, Inception V3, and DenseNet-121) along with three classifiers here and their performances are shown in Table 5.

Table 3
Classification performance (in %) of different CNN architectures with classifiers on PH2 dataset.

Network	Classifier	Accuracy	Precision	Sensitivity	Specificity	F1-score
AlexNet	DT	76.66	37.50	25.00	89.58	30.00
	LDA	83.33	58.33	58.33	89.58	58.33
	AB	85.00	63.63	58.33	91.66	60.86
	RF	86.66	70.00	58.33	93.75	63.63
	KNN	86.66	100.00	33.33	100.00	50.00
	LR	80.00	0.00	0.00	100.00	0.00
	SVM Linear	80.00	0.00	0.00	100.00	0.00
	SVM RBF	85.00	100.00	25.00	100.00	40.00
	NB	81.66	53.33	66.66	85.41	59.25
	MLP	**88.33**	77.77	**58.33**	95.83	**66.66**
VGG-16	DT	88.33	85.71	50.00	97.91	63.15
	LDA	86.66	75.00	50.00	95.83	60.00
	AB	90.00	87.50	58.33	97.91	70.00
	RF	90.00	100.00	50.00	100.00	66.66
	KNN	86.66	83.33	41.66	97.91	55.55
	LR	93.33	100.00	66.66	100.00	80.00
	SVM Linear	**93.33**	100.00	**66.66**	100.00	**80.00**
	SVM RBF	85.00	63.63	58.33	91.66	60.86
	NB	86.66	64.28	75.00	89.58	69.23
	MLP	90.00	80.00	66.66	95.83	72.72
VGG-19	DT	76.66	0.00	0.00	95.83	0.00
	LDA	83.33	58.33	58.33	89.58	58.33
	AB	90.00	100.00	50.00	100.00	66.66
	RF	88.33	85.71	50.00	97.91	63.15
	KNN	88.33	100.00	41.66	100.00	58.82
	LR	91.66	100.00	58.33	100.00	73.68
	SVM Linear	86.66	100.00	33.33	100.00	50.00
	SVM RBF	86.66	83.33	41.66	97.91	55.55
	NB	**95.00**	90.90	**83.33**	97.91	**86.95**
	MLP	90.00	87.50	58.33	97.91	70.00
Inception V3	DT	78.33	45.45	41.66	87.50	43.47
	LDA	90.00	80.00	66.66	95.83	72.72
	AB	91.66	81.81	75.00	95.83	78.26
	RF	88.33	100.00	41.66	100.00	58.82
	KNN	90.00	87.50	58.33	97.91	70.00
	LR	**95.00**	90.90	**83.33**	97.91	**86.95**
	SVM Linear	93.33	90.00	75.00	97.91	81.81
	SVM RBF	90.00	80.00	66.66	95.83	72.72
	NB	91.66	76.92	83.33	93.75	80.00
	MLP	88.33	69.23	75.00	91.66	72.00
ResNet-50	DT	**86.66**	70.00	**58.33**	93.75	**63.63**
	LDA	83.33	66.66	33.33	95.83	44.44
	AB	85.00	80.00	33.33	97.91	47.05
	RF	80.00	50.00	16.66	95.83	25.00
	KNN	78.33	0.00	0.00	97.91	0.00
	LR	85.00	80.00	33.33	97.91	47.05
	SVM Linear	80.00	0.00	0.00	100.00	0.00
	SVM RBF	80.00	0.00	0.00	100.00	0.00
	NB	58.33	27.58	66.66	56.25	39.02
	MLP	85.00	80.00	33.33	97.91	47.05
MobileNet	DT	68.33	33.33	58.33	70.83	42.42
	LDA	90.00	80.00	66.66	95.83	72.72
	AB	93.33	90.00	75.00	97.91	81.81
	RF	88.33	85.71	50.00	97.91	63.15
	KNN	**93.33**	83.33	**83.33**	95.83	**83.33**
	LR	91.66	88.88	66.66	97.91	76.19
	SVM Linear	91.66	88.88	66.66	97.91	76.19
	SVM RBF	91.66	81.81	75.00	95.83	78.26
	NB	91.66	81.81	75.00	95.83	78.26
	MLP	88.33	72.72	66.66	93.75	69.56
EfficientNet B0	DT	81.66	100.00	8.33	100.00	15.38
	LDA	83.33	58.33	58.33	89.58	58.33
	AB	85.00	71.42	41.66	95.83	52.63
	RF	81.66	55.55	41.66	91.66	47.61
	KNN	83.33	62.50	41.66	93.75	50.00
	LR	80.00	0.00	0.00	100.00	0.00
	SVM Linear	80.00	0.00	0.00	100.00	0.00
	SVM RBF	80.00	0.00	0.00	100.00	0.00
	NB	30.00	21.15	91.66	14.58	34.37
	MLP	**90.00**	87.50	**58.33**	97.91	**70.00**
DenseNet-121	DT	78.33	47.36	75.00	79.16	58.06
	LDA	86.66	66.66	66.66	91.66	66.66
	AB	91.66	76.92	83.33	93.75	80.00
	RF	91.66	100.00	58.33	100.00	73.68
	KNN	91.66	81.81	75.00	95.83	78.26
	LR	93.33	90.00	75.00	97.91	81.81
	SVM Linear	93.33	90.00	75.00	97.91	81.81
	SVM RBF	93.33	83.33	83.33	95.83	83.33
	NB	93.33	75.00	100.00	91.66	85.71
	MLP	**98.33**	**92.30**	**100.00**	**97.91**	**96.00**

(*continued on next page*)

Table 3 (*continued*).

The results shown in bold represents the best performance on a particular train-test scenario. The DenseNet-121 with MLP classifier performed better compared to VGG-16 and Inception V3 in almost all scenarios. It can also be noticed that the reduction of training data does not decrease the melanoma classification performance significantly.

We also investigated the classification performance of DenseNet-121 with ten different classifiers in terms of accuracy, sensitivity, and F1-score. The results are depicted in Fig. 5. It can be observed that MLP classifier achieved higher or comparable accuracy in all the scenarios.

5.2.3. Experiments on different normalization techniques

The aim of this experiment is to test the impact of image normalization techniques on melanoma classification. Here, we explored four image normalization techniques: all-image-norm, per-image-norm, per-channel-norm, and max-pixel-norm and the results of the experiment with the best performing approach are shown in Table 6.

It is evident that the DenseNet121+MLP approach achieved better melanoma classification performance while using per-channel-norm normalization technique compared to other normalization techniques. The per-channel-norm technique obtained 98.33% of accuracy which is ∼1.67%, ∼3.33% and ∼13.33% higher than the per-image-norm, all-image-norm and max-pixel-norm, respectively. The model achieved 100% of sensitivity with per-channel-norm which is ∼8.34% higher than the per-image-norm, all-image-norm and ∼50% higher than the max-pixel-norm technique; whereas, the F1-score is ∼4.34%, ∼8% and ∼38.86% higher than per-image-norm, all-image-norm and max-pixel-norm, respectively.

For normalization with per-image-norm normalization, the model obtained a comparable performance. This experiment demonstrated that the per-channel-norm normalization technique achieved superior performance with DenseNet-121+MLP compared to other normalization techniques.

5.2.4. Experiments on CNN features at different layers

The size of the feature set varies if we extract the features from different CNN layers. In this experiment, we investigate the effect of such deep features with three best performing CNN architectures and classifiers. For this, we derived features from five different layers of these CNN models and computed their performances independently. Also, we tested the performance using the fusion of the features of the five layers. It is worth noting here that these features are fused using concatenation. The results for DenseNet-121, Inception V3, and VGG-16 are depicted in Figs. 6, 7, and 8.

As shown in Figs. 6 and 8, the last layer features for DenseNet-121 (1024) and VGG-16 (4096) result in higher classification performance compared to other layer and fused features. For Inception V3, the intermediate layer features (12288) obtained better results compared to other layer and fused features. However, a comparable performance was achieved with last layer features (2048). It can be observed that the performance is improved when we derive high-level features compared to low-level features.

We also evaluated the performance of the best performing models with three different feature fusion techniques. The first one is early-stage fusion also referred to as data-level fusion or input-level fusion

Table 4
Classification results (in %) with and without preprocessing.

Method	Preprocessing	Accuracy	Precision	Sensitivity	Specificity	F1-score
DenseNet-121 + MLP	Without	95.00	90.90	83.33	97.91	86.95
	With	**98.33**	**92.30**	**100.00**	**97.91**	**96.00**

Table 5
Classification results (in %) of best performing methods with different train-test split scenarios.

Network	Classifier	Acc	Sen	F1	Acc	Sen	F1	Acc	Sen	F1	Acc	Sen	F1	Acc	Sen	F1
		80:20 (train-test)%			**70:30 (train-test)%**			**60:40 (train-test)%**			**50:50 (train-test)%**			**5-Fold cross validation(%)**		
VGG-16	LR	92.50	66.66	80.00	93.33	66.66	80.00	92.50	64.70	78.57	89.00	47.36	62.06	88.50	57.50	66.66
	SVM Linear	92.50	66.66	80.00	93.33	66.66	80.00	88.75	47.05	64.00	89.00	42.10	59.25	90.00	57.50	69.69
	MLP	90.00	66.66	75.00	90.00	66.66	72.72	**93.75**	**82.35**	**84.84**	89.00	63.15	68.57	88.50	60.00	67.60
Inception V3	LR	85.00	33.33	50.00	95.00	83.33	86.95	86.25	47.05	59.25	89.00	47.36	62.06	86.00	52.50	60.00
	SVM Linear	87.50	44.44	61.53	93.33	75.00	81.81	88.75	64.70	70.96	89.00	57.89	66.66	87.00	57.50	63.88
	MLP	87.50	44.44	61.53	88.33	75.00	72.00	83.75	47.05	55.17	88.00	52.63	62.50	84.00	52.50	56.75
DenseNet-121	LR	92.50	77.77	82.35	93.33	75.00	81.81	92.50	70.58	80.00	93.00	68.42	78.78	**91.00**	**72.50**	**76.31**
	SVM Linear	92.50	77.77	82.35	93.33	75.00	81.81	90.00	70.58	75.00	91.00	68.42	74.28	90.50	70.00	74.66
	MLP	**97.50**	**100.00**	**94.73**	**98.33**	**100.00**	**96.00**	92.50	82.35	82.35	**93.00**	**73.68**	**80.00**	90.00	75.00	75.00

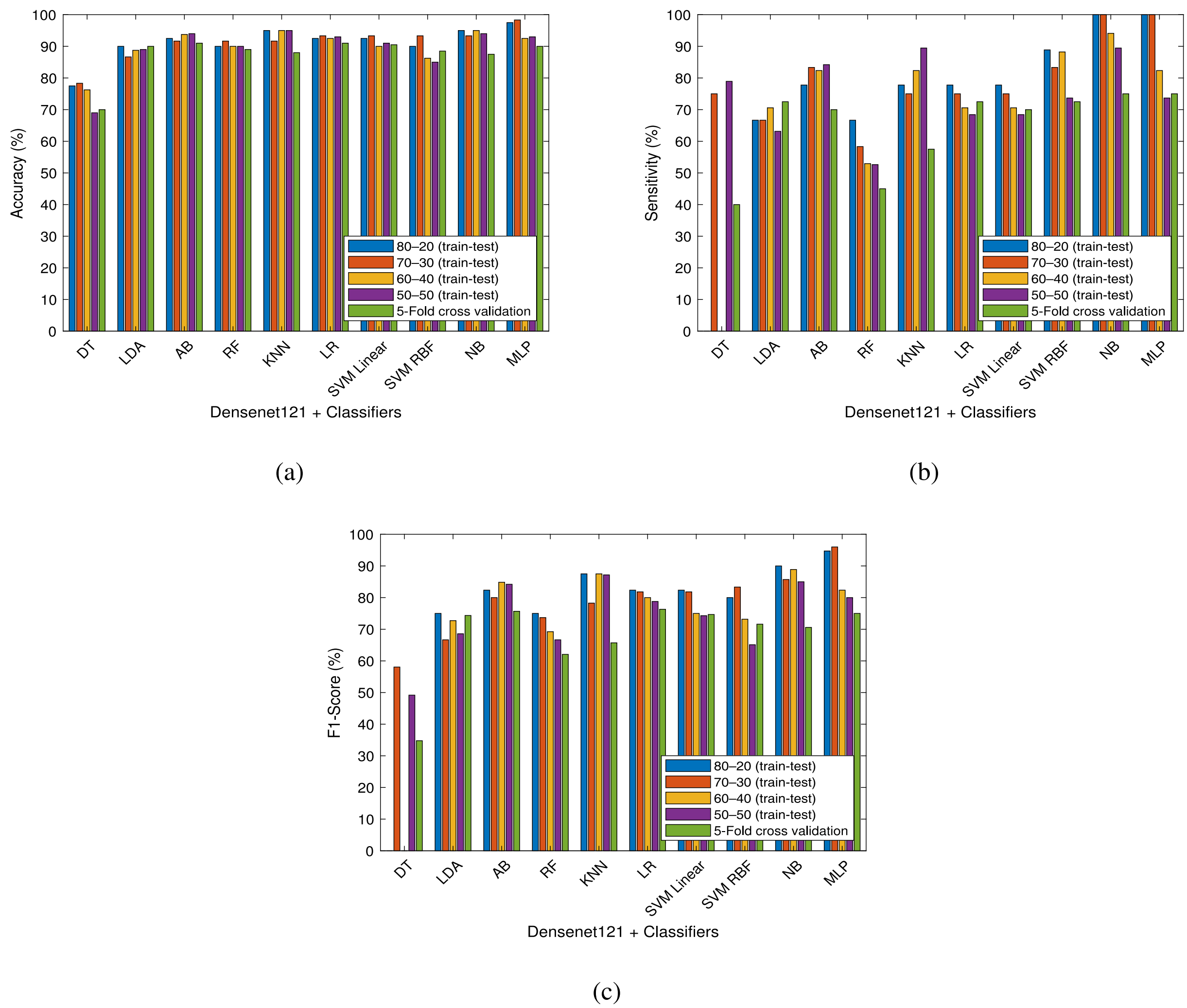

Fig. 5. Classification performance of DenseNet-121 with different classifiers in terms of accuracy, sensitivity and F1-score under the train-test splitting scenarios.

where the features at the initial layers were concatenated. We concatenated features from two such layers, resulting 1204224, 1085472 and 1204224 number of features for DenseNet-121, Inception V3 and VGG-16, respectively. Secondly, we concatenated features at the intermediate layers which is known as intermediate-level fusion. The intermediate-level fusion helps in transforming the input data into high-level features through multiple layers. Here, the features at two intermediate layers are fused which results in 200704, 67776 and 125440 features for DenseNet-121, Inception V3 and VGG-16, respectively. Finally, we concatenated the features derived from five different layers including the last layer and this is referred to as decision-level fusion. Using this fusion, we derived 1456128, 1155296 and 1333760

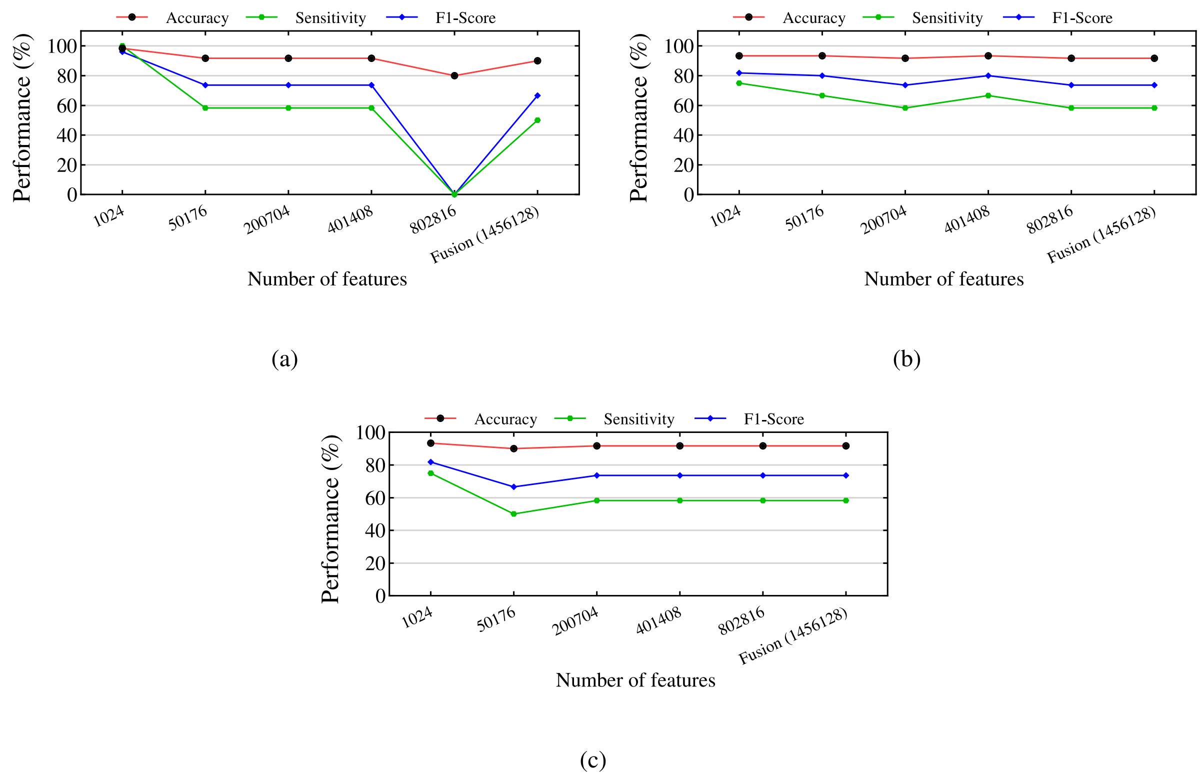

Fig. 6. Classification performance of deep features extracted from different layers of DenseNet-121 with different classifiers: (a) MLP, (b) SVM Linear, and (c) LR.

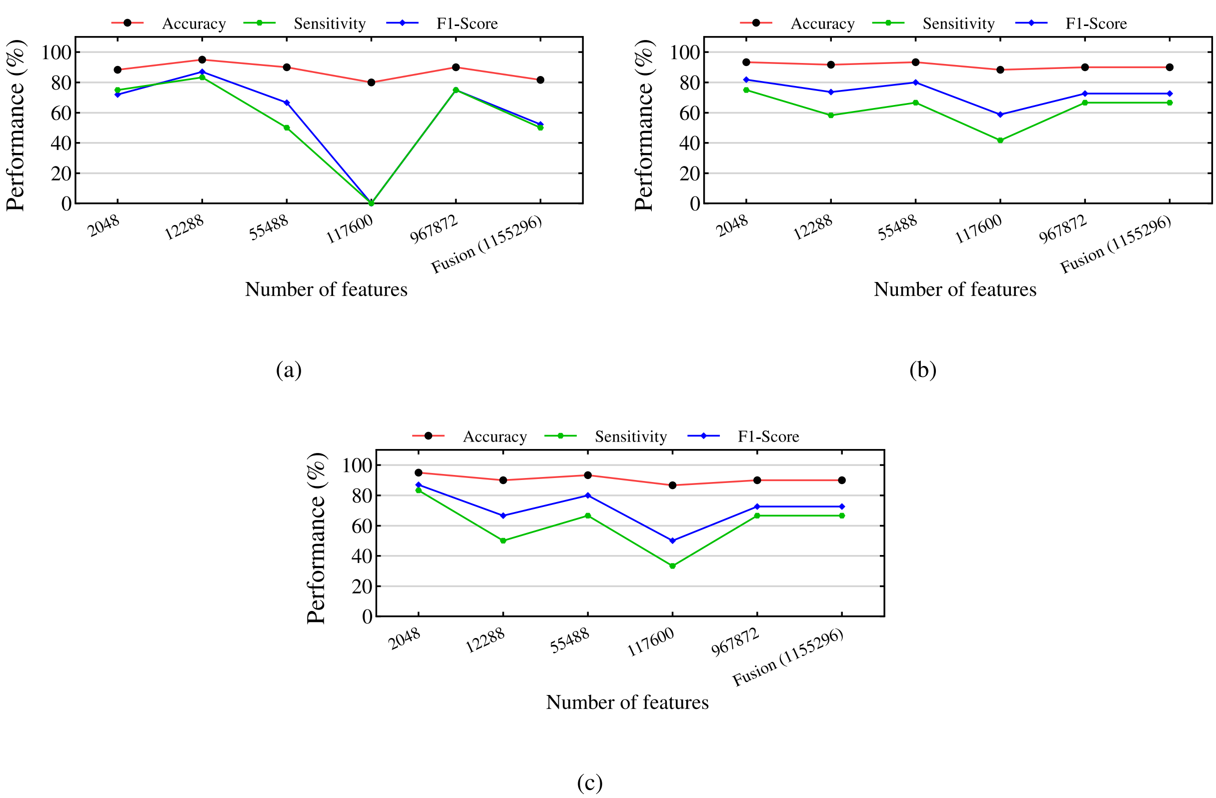

Fig. 7. Classification performance of deep features extracted from different layers of Inception V3 with different classifiers: (a) MLP, (b) SVM Linear, and (c) LR.

features for DenseNet-121, Inception V3 and VGG-16, respectively. The detailed results for each fusion type are shown in Table 7. It can be observed here that the performance of each model using different fusion techniques is comparatively lower than the performance achieved only using the last layer features.

5.3. Comparison of proposed method with other approaches on PH^2 dataset

Table 8 summarizes the performance of the best performing approach 'DenseNet-121+MLP' with other existing melanoma classification methods on PH^2 dataset. The JRC techniques achieved 92% of

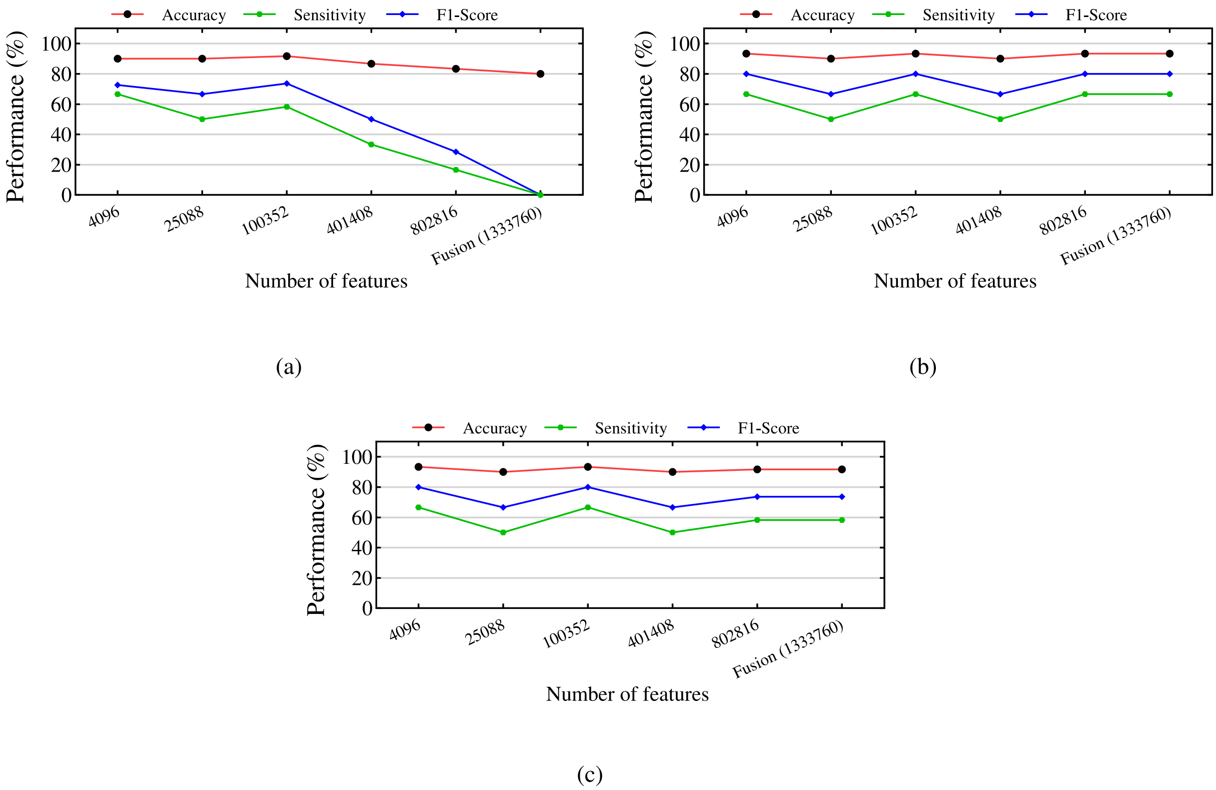

Fig. 8. Classification performance of deep features extracted from different layers of VGG-16 with different classifiers: (a) MLP, (b) SVM Linear, and (c) LR.

accuracy, whereas ANN and VGG+LR achieved 92.5% of accuracy. The combination of Autoencoder and SVM performed well with 94% of accuracy. Our recent method MobileNet+SVM achieved 95% of accuracy. In contrast, DenseNet-121+MLP obtained the highest classification accuracy of 98.33% compared to state-of-the-art methods. The bag tree ensemble classifier achieved 96% of sensitivity among compare methods but, DenseNet-121+MLP achieved 100% of sensitivity and that increase trust on our method for melanoma classification. The proposed method derived results by preparing a dataset by ROI extraction with boundary localization, applying rectangle crop and getting only interested region of skin lesion, and an per-channel-norm normalization strategy. Also, DenseNet-121 extract valuable features of skin lesions because of deep convolutional. It proves that a combination of CNN and classifiers promise the best results compared to others.

5.4. Comparison of proposed method with other approaches on ISIC 2016 dataset

Table 9 shows the performance of DenseNet-121 with different classifiers on ISIC 2016 dataset. The classifier SVM with DenseNet-121 achieved higher accuracy compared to others, whereas MLP achieved a higher F-score value.

The comparison of the DenseNet-121+MLP approach with other CNN models in terms of sensitivity on the ISIC 2016 dataset is shown in Fig. 9. As shown, the DenseNet-121+MLP achieved a higher sensitivity of 56% compared to other models, whereas the AlexNet+SVM achieved a 53.3% of sensitivity.

5.5. Results of proposed method on ISIC 2017 and HAM10000 datasets with other CNN models

Table 10 shows the performance of the three best performing models such as VGG16, Inception V3 and DenseNet-121 on ISIC 2017 and HAM10000 datasets. It can be observed that DenseNet-121 achieved higher accuracy of 82.83% and 81.00% on ISIC 2017 and HAM10000 datasets, respectively.

Also, it can be noticed that the performance of the selected networks on the HAM10000 dataset is relatively low. This is due to the fact that HAM10000 consists of images of seven different classes and is highly imbalanced. Further, it is worth noting that the boundary localization has not been adopted over HAM10000 due to the unavailability of image masks. Also, note that other influencing parameters have been kept similar for a fair comparison.

6. Conclusions

In this paper, we analyzed the dermoscopy images for the diagnosis of melanoma via deep features extracted from a variety of pre-trained CNN architectures. We suggested to employ boundary localization that helps in preserving essential skin lesion regions and thereby, facilitating the extraction of more prominent features. Then, we explored the effectiveness of eight different pre-trained CNN models for deep feature extraction from those regions and employed a set of classifiers to detect melanoma. We carried out an extensive set of experiments using four standard datasets and testified the influence of numerous crucial factors such as preprocessing, train-test split ratio, normalization technique, and features at different CNN layers for melanoma classification. Also, we verified the performance with various feature fusion techniques. The results demonstrate that 'DenseNet-121+MLP' achieved higher classification performance on both datasets compared to other combinations of methods as well as state-of-the-art approaches. We hope our investigations in this paper will serve as additional material for the readers to design more effective models using deep features for melanoma classification. Our future studies include testing the proposed approach on large and diverse datasets, developing an ensemble CNN model, and encouraging its use in clinical practice.

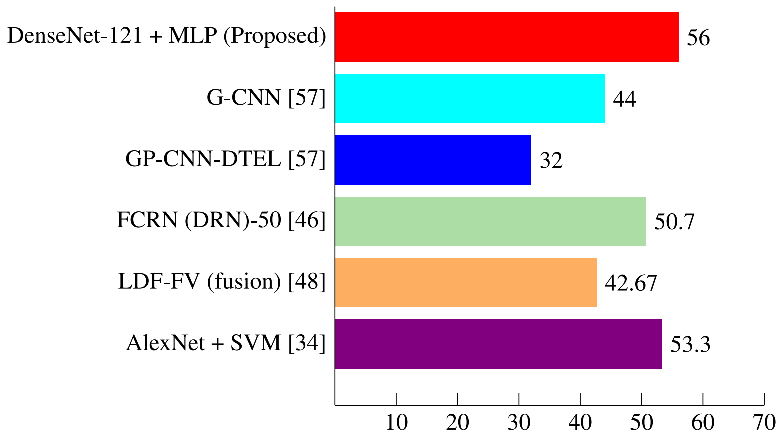

Fig. 9. Performance comparison (in %) with existing methods on ISIC 2016 dataset.

Table 6
Impact of various image normalization techniques (in %) with DenseNet-121 and different classifiers.

Classifier	Image-normalization	Accuracy	Sensitivity	F1-score
DT	max-pixel-norm	80.00	0.00	0.00
	all-image-norm	**91.66**	66.66	76.19
	per-image-norm	81.66	75.00	62.06
	per-channel-norm	78.33	75.00	58.06
LDA	max-pixel-norm	**93.33**	75.00	81.81
	all-image-norm	86.66	58.33	63.63
	per-image-norm	86.66	66.66	66.66
	per-channel-norm	86.66	66.66	66.66
AB	max-pixel-norm	93.33	66.66	80.00
	all-image-norm	**93.33**	66.66	80.00
	per-image-norm	91.66	66.66	76.19
	per-channel-norm	91.66	83.33	80.00
RF	max-pixel-norm	90.00	50.00	66.66
	all-image-norm	**93.33**	66.66	80.00
	per-image-norm	93.33	66.66	80.00
	per-channel-norm	91.66	58.33	73.68
KNN	max-pixel-norm	85.00	41.66	52.63
	all-image-norm	91.66	58.33	73.68
	per-image-norm	90.00	50.00	66.66
	per-channel-norm	**91.66**	75.00	78.26
LR	max-pixel-norm	88.33	50.00	63.15
	all-image-norm	83.33	50.00	54.54
	per-image-norm	83.33	50.00	54.54
	per-channel-norm	**93.33**	75.00	81.81
SVM Linear	max-pixel-norm	88.33	50.00	63.15
	all-image-norm	86.66	58.33	63.63
	per-image-norm	86.66	58.33	63.63
	per-channel-norm	**93.33**	75.00	81.81
SVM RBF	max-pixel-norm	88.33	58.33	66.66
	all-image-norm	90.00	66.66	72.72
	per-image-norm	88.33	66.66	69.56
	per-channel-norm	**93.33**	83.33	83.33
NB	max-pixel-norm	83.33	75.00	64.28
	all-image-norm	**96.66**	100.00	92.30
	per-image-norm	95.00	100.00	88.88
	per-channel-norm	93.33	100.00	85.71
MLP	max-pixel-norm	85.00	50.00	57.14
	all-image-norm	95.00	91.66	88.00
	per-image-norm	96.66	91.66	91.66
	per-channel-norm	**98.33**	**100.00**	**96.00**

Table 7
Performance evaluation (in %) of different feature fusion techniques.

Network	Classifier	Types of fusion	Accuracy	Sensitivity	F1-score
VGG-16	LR	Data level	91.66	58.33	73.68
		Intermediate level	93.33	66.66	80.00
		Decision level	91.66	58.33	73.68
	SVM Linear	Data level	93.33	66.66	80.00
		Intermediate level	93.33	66.66	80.00
		Decision level	93.33	66.66	80.00
	MLP	Data level	80.00	0.00	0.00
		Intermediate level	**93.33**	66.66	80.00
		Decision level	80.00	0.00	0.00
Inception V3	LR	Data level	90.00	66.66	72.72
		Intermediate level	93.33	66.66	80.00
		Decision level	90.00	66.66	72.72
	SVM Linear	Data level	90.00	66.66	72.72
		Intermediate level	93.33	66.66	80.00
		Decision level	90.00	66.66	72.72
	MLP	Data level	78.33	16.66	23.52
		Intermediate level	**93.33**	66.66	80.00
		Decision level	81.66	50.00	52.17
DenseNet-121	LR	Data level	91.66	58.33	73.68
		Intermediate level	91.66	58.33	73.68
		Decision level	91.66	58.33	73.68
	SVM Linear	Data level	91.66	58.33	73.68
		Intermediate level	91.66	58.33	73.68
		Decision level	91.66	58.33	73.68
	MLP	Data level	90.00	50.00	66.66
		Intermediate level	**91.66**	58.33	73.68
		Decision level	90.00	50.00	66.66

CRediT authorship contribution statement

Himanshu K. Gajera: Conceptualization, Methodology, Software, Validation, Visualization, Writing – original draft, Data curation, Formal analysis. **Deepak Ranjan Nayak:** Conceptualization, Methodology, Writing – review & editing, Validation, Supervision. **Mukesh A. Zaveri:** Writing – review & editing, Validation, Supervision.

Declaration of competing interest

No author associated with this paper has disclosed any potential or pertinent conflicts which may be perceived to have impending conflict with this work. For full disclosure statements refer to https://doi.org/10.1016/j.bspc.2022.104186.

Data availability

The datasets are publicly available.

Table 8
Comparison of proposed method with existing approaches on PH^2 dataset.

Method	Year	Performance (%)		
		Accuracy	Sensitivity	Specificity
SIFT features + JRC [43]	2016	92.00	87.50	93.13
Hand-designed features + ANN [41]	2017	92.50	90.86	96.11
Modified AlexNet [86]	2018	80.00	72.92	83.33
VGG-19 + LR [65]	2018	92.50	75.00	96.88
AlexNet [87]	2018	93.00	86.00	94.00
ABCD features + Bag tree ensemble classifier [88]	2019	93.50	96.00	93.00
ABCD features + SVM (RBF) [89]	2019	88.00	80.00	90.00
Autoencoder + SVM [49]	2020	94.00	90.00	66.66
MobileNet + SVM [55]	2021	95.00	83.33	97.91
DenseNet-121 + MLP (Proposed)		**98.33**	**100**	**97.91**

Table 9
Performance evaluation (in %) of DenseNet-121 with other classifiers on ISIC 2016 dataset.

Classifier	Accuracy	Precision	Sensitivity	Specificity	F1-score
DT	80.21	0.00	0.00	100.00	0.00
LDA	65.69	28.00	46.66	70.39	35.00
AB	76.78	38.59	29.33	88.48	33.33
RF	81.79	75.00	12.00	99.01	20.68
KNN	77.83	39.53	22.66	91.44	28.81
LR	82.32	58.00	38.66	93.09	46.40
SVM Linear	**82.84**	66.66	26.66	96.71	38.09
SVM RBF	81.79	56.81	33.33	93.75	42.01
NB	70.97	36.22	**61.33**	73.33	45.54
MLP	**80.47**	50.60	**56.00**	86.51	**53.16**

Table 10
Results of proposed method on ISIC 2017 and HAM10000 datasets.

Network	Classifier	Accuracy (%)	
		ISIC 2017	HAM10000
VGG16	LR	82.00	76.00
	SVM Linear	80.50	73.00
	MLP	78.16	75.00
Inception V3	LR	81.16	78.00
	SVM Linear	80.66	74.00
	MLP	78.66	77.00
DenseNet-121	LR	**82.83**	79.00
	SVM Linear	81.66	79.00
	MLP	**81.16**	**81.00**

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